

Comparing social needs for transport, gaps in transit service and the Mobility Index for the west of Ireland

Dr James Reynolds

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Introduction

Research Background

A wide selection of indexes, metrics and other indicators have been developed by researchers, practitioners and others to assess transport and related factors. Examples include: *Walk Score* and *Transit Score*,¹ which rate a location out of 100 for walkability and transit accessibility respectively; metrics within the *Transit Capacity and Quality of Service Manual (TCQSM)*;² and the UK's Public Transport Accessibility Level (PTAL) approach and the Public Transport Accessibility Index (PTAI)³

What may sometimes be lacking is comparison of the results obtained using one set of indicators versus others. In part this can be because of the variability in data availability. For example, national censuses often provide a key resource for understanding transport-related issues, yet the collection practices used, questions asked and other characteristics tend to be highly variable from one jurisdiction to another. As well, some indicators have been developed for specific contexts, for example McHenry et al. (2023)⁴ recently developed a Mobility Index (MI) for rural centres in the west of Ireland/ This differs from other MIs, which may tend to be focused on urbanised locations.

Finally, some indicators are difficult or impossible to calculate independently. Some are closed-source, in that they may be proprietary or otherwise only available from a single website⁵. Others may just be challenging to calculate due to a lack of tools. Some research progress is being made on this, however, with (for example) James Wong⁶ reporting the development of a set of software tools that allow many of the TCQSM indicators to be calculated from General Transit Feed Specification (GTFS) datasets. Wong⁷ shows results comparing 50 USA transit operators, but has also published the code base (on github) to allow others to undertake similar comparisons in other places or at other times.

While censuses and the collection of many other datasets at local, regional and national levels appear likely to remain specific to their own geographic contexts, there have been some recent developments in technology and transport practices that allow for universal (or at

¹ Walk Score, "Transit Score Methodology," 2023, <https://www.walkscore.com/transit-score-methodology.shtml>.

² Kittleson & Associates et al., *Transit Capacity and Quality of Service Manual, Third Edition*, Third Edition, TCRP Report 165 (Washington DC: Transportation Research Board; Transportation Research Board, 2013), <http://www.trb.org/Main/Blurbs/169437.aspx>.

³ E Saghapour, S Moridpour, and R Thompson, "Public Transport Accessibility in Metropolitan Areas: A New Approach Incorporating Population Density," *Journal of Transport Geography* 54 (2016): 273–85.

⁴ Helen McHenry, Amaya Vega, and Catherine Swift, "Understanding Mobility in Rural Towns: Development of a Mobility Index for the West of Ireland," *Transportation Research Procedia* 72 (2023): 3730–37, doi:<https://doi.org/10.1016/j.trpro.2023.11.545>.

⁵ e.g. WalkScore and Transit Score cannot be independently calculated, and the exact algorithm remains unpublished

⁶ "Leveraging the General Transit Feed Specification for Efficient Transit Analysis," *Transportation Research Record* 1, no. 2338 (2013): 11–19, doi:10.3141/2338-02.

⁷ Ibid.

least internationally-applicable) tools and approaches. The aforementioned GTFS allows transit operators to publish their static timetable data in an open, simple and text-based format. This was originally developed to support the provision of transit directions in Google Maps, but has since been adopted by over 10,000 agencies.⁸ Hence, software tools that take GTFS data as inputs can generally be adapted easily and rapidly to many systems and locations.

One such recently developed software tool is the *gtfssupplyindex* R package,⁹ which builds on other GTFS-related R packages including *gtfstools*;¹⁰ and *tidytransit*.¹¹ The *gtfssupplyindex* can be used to calculate a transit Supply Index, developed by Graham Currie et al.¹² Graham Currie,¹³ Graham Currie and Zed Senbergs¹⁴ and Graham Currie¹⁵ in research related to identifying gaps in public transport services on the basis of social needs for transport. This approach, and its potential application to route and service planning, and for policy making was suggested to be “substantially more useful than the presentation of anecdotal evidence, which is the most common means of identifying transport needs in local transport studies throughout the world”.¹⁶ However, its development predated the GTFS, and the results presented in Currie¹⁷ for Melbourne, Australia, in 2006 had required bespoke cleaning and processing of data obtained directly from the transit agency.

With the *gtfssupplyindex* package, however, results can be obtained directly from GTFS files. However, this package and approach has yet to be applied outside of the urban and suburban areas of Melbourne (as reported in Reynolds, Currie, and Qu¹⁸). It is unclear whether transit Supply Index and needs-gap assessment approach can be easily applied to other jurisdictions or to more rural locations. Also unclear is how such results might compare to other transport-related indicators, including those developed for rural areas (such as the McHenry, Vega, and Swift¹⁹ MI).

In the following sections, therefore, this paper reports the application of the *gtfssupplyindex* package to the west of Ireland, adopting the same case and context as McHenry, Vega, and Swift.²⁰ Results from a needs-gap analysis are compared to those from McHenry, Vega, and Swift²¹ so as to provide insights into the differences between these indicators. This research is anticipated to be part of a larger programme that would continue to apply different indicators and approaches to this case, so as to allow deeper understandings of both transport in the west of Ireland, but also how well existing techniques might help researchers, practitioners and policy-makers understand and improve networks.

⁸ MobilityData, *General Transit Feed Specification (GTFS)*, undated, %7Bhttps://gtfs.org/%7D.

⁹ James Reynolds, Graham Currie, and Yanda Qu, “Social Needs for Transport and Gaps in Transit Service: New GTFS Tools,” *Submitted to Australasian Transport Research Forum*. November, Auckland, NZ., under review.

¹⁰ **R-gtfstools?**

¹¹ **R-tidytransit?**

¹² “Quantitative Approaches to Needs Based Assessment of Public Transport Services: The Hobart Transport Needs Gap Study,” Journal Article, 2003, <https://www.semanticscholar.org/paper/2c049091cafb56c66efc532ad2bdd774d8efc0eb>.

¹³ “Gap Analysis of Public Transport Needs: measuring Spatial Distribution of Public Transport Needs and Identifying Gaps in the Quality of Public Transport Provision,” *Transportation Research Record* 1895 (2004): 137–46, doi:10.3141/1895-18.

¹⁴ “Identifying Spatial Gaps in Public Transport Provision for Socially Disadvantaged Australians: The Melbourne ‘Needs Gap’ Study,” 2007.

¹⁵ “Quantifying Spatial Gaps in Public Transport Supply Based on Social Needs,” *Journal of Transport Geography* 18, no. 1 (2010): 31–41.

¹⁶ Ibid.

¹⁷ Ibid.

¹⁸ “Social Needs for Transport and Gaps in Transit Service.”

¹⁹ “Understanding Mobility in Rural Towns.”

²⁰ Ibid.

²¹ Ibid.

Methodology

The transit Supply Index (SI)

Reynolds, Currie, and Qu²² provide a generalized form of the Currie²³ SI equation, which is:

$$SI_{area,time} = \sum_{i=1}^n \frac{Area_{Bi}}{Area_{area}} SL_{i,time}$$

where:

- $SI_{area,time}$ is the Supply Index for the *area of interest* and a given period of time;
- $Area_{Bi}$ is the amount of the buffer zone for each stop (i) that is within the *area of interest*²⁴;
- $Area_{area}$ is the area of each *area of interest*; and
- $SL_{i,time}$ is the number of transit arrivals for each stop (i) within the given time period.

Based on the SI scores Currie²⁵ grouped areas of interest into seven categories. The first category includes areas of interest with zero SI scores, i.e. those areas without any transit services at all. The remaining areas of interest are then categorised into three even groups (Very Low, Low and Below Average) across those with areas with SI scores below average, and three even groups above the average²⁶ (Above Average, High and Very High). This approach, which was also used in Reynolds, Currie, and Qu,²⁷ has been used in this paper as well so as to maintain consistency with the previous research.

The areas of interest adopted in this study at the 2022 definition of *Small Areas*, used in the Irish Census. However, data from the Irish Census is reported separately for Galway City and the rest of County Galway²⁸, whereas in this analysis Small Areas within Galway City are reported as part of County Galway

GTFS data

Data for this paper was sourced from Transport for Ireland²⁹ on October 1, 2025³⁰. The analysis was run for the week starting October 1.

²² "Social Needs for Transport and Gaps in Transit Service."

²³ "Quantifying Spatial Gaps in Public Transport Supply Based on Social Needs."

²⁴ In Currie, *ibid* this buffer zone was based on a radius of 400 metres for bus and tram stops, and 800 metres for railway stations. The same definition is used here.

²⁵ *Ibid*.

²⁶ Based on only including areas of interest with non-zero scores

²⁷ "Social Needs for Transport and Gaps in Transit Service."

²⁸ "Galway City" is included as an additional county in the COUNTY_ENGLISH field

²⁹ "Public Transport Data" (National Transport Authority, undated), https://www.transportforireland.ie/transitData/PT_Data.html.

³⁰ Selecting the "All GTFS Operators" feed

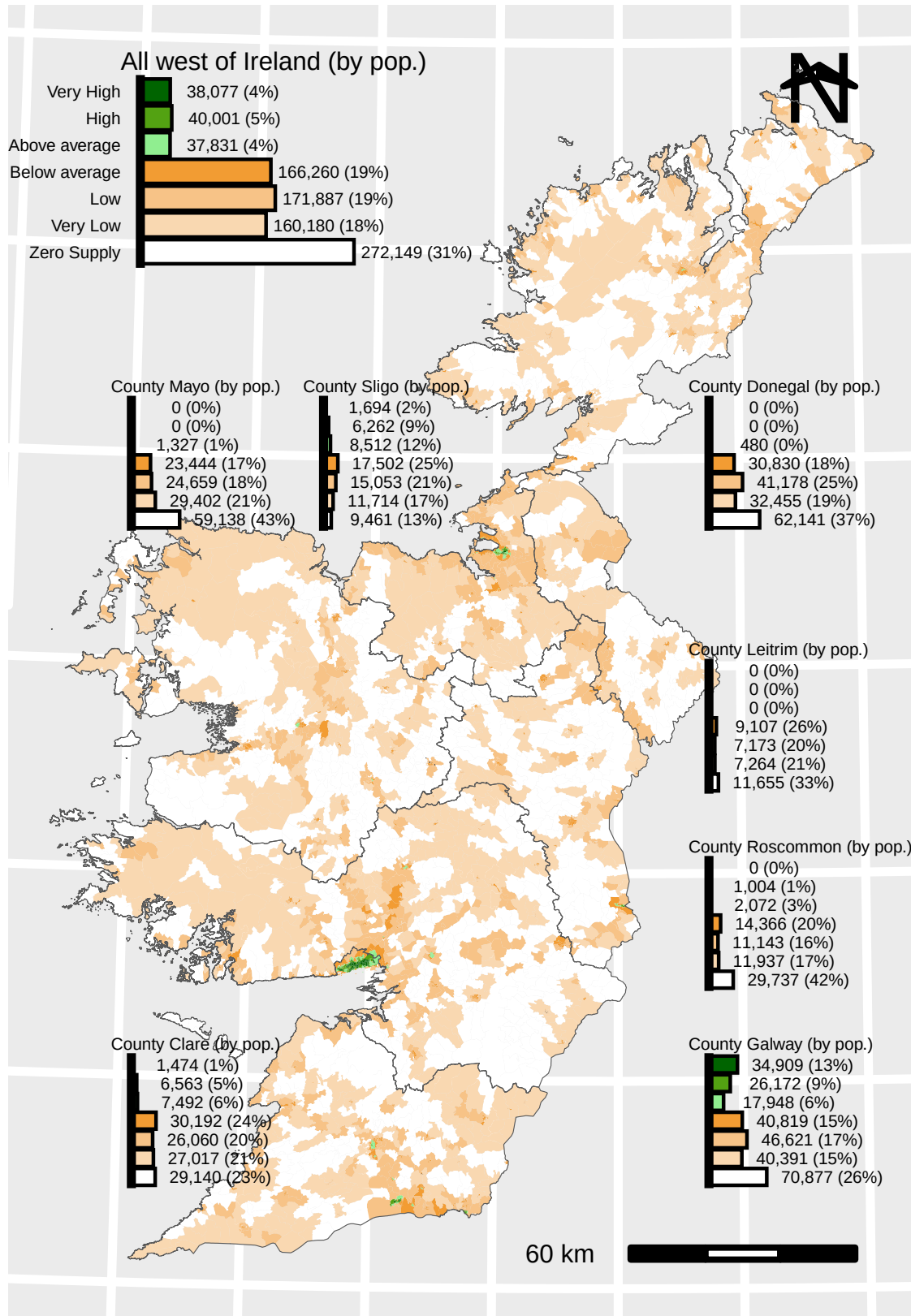


Figure 1: Transit Supply Index categories across the west of Ireland, by Small Area and by population

Characteristic	Overall N = 3,666	Transit supply category						
		Zero Supply N = 1,172	Very Low N = 683	Low N = 683	Below average N = 683	Above average N = 149	High N = 148	Very High N = 148
SI								
Min	0.0	0.0	0.0	2.3	37.3	418.5	986.6	2,178.9
Q1	0.0	0.0	0.1	4.6	64.2	513.1	1,286.1	2,610.6
Mean	281.1	0.0	0.7	13.6	144.3	665.4	1,586.6	3,973.2
Q3	73.0	0.0	1.2	21.3	197.1	812.9	1,951.6	4,140.4
Max	11,860.5	0.0	2.3	37.2	411.3	978.1	2,177.0	11,860.5
Sum	1,030,354.4	0.0	507.8	9,282.6	98,566.1	99,148.7	234,815.0	588,034.3
population								
Min	69	71	74	93	69	99	83	78
Q1	183	175	178	193	181	199	220	194
Mean	242	232	235	252	243	254	270	257
Q3	289	280	284	297	295	288	311	301
Max	827	577	669	760	827	581	537	785
Sum	886,385	272,149	160,180	171,887	166,260	37,831	40,001	38,077

Results

All of the west of Ireland

Table 1 shows summary statistics for SI scores and the population, overall for all of the west of Ireland and split by transit supply category. The total population of the west of Ireland is 886,385, of whom 87% (770,476) live in Small Areas where the SI score is below the average (of those Small Areas with a non-zero SI score) (413.1). This includes the 272,149 people (31%) who live in Small Areas with no transit at all.

Figure 1 shows transit Supply Index (SI) categories by Small Areas across the west of Ireland. It also shows to the population living in Small Areas in each category for both the entire west of Ireland and for each county. In general, there appear to be large areas of the west of Ireland with SI scores below the average, or with no transit supply at all. Also, as might be expected, Small Areas where SI scores are higher than the average (of those with non-zero scores) appears to be located in very small and scattered geographic areas (at population centers).

Table 2 summarises the Small Area populations in each County by transit supply category, shown as a share of the total west of Ireland population. Table 3 shows SI scores and transit supply categories by County, weighted by population.

Table 1: Summary statistics; SI score by transit supply category, by Small Area.

Table 1: West of Ireland: County Small Area populations by Transit Supply grouping

County	Transit Supply category							Total
	Zero Supply	Very Low	Low	Below av.	Above av.	High	Very High	

Table 2: County Small Area populations, by transit supply category.

GALWAY	8.0% (70,877)	4.6% (40,391)	5.3% (46,621)	4.6% (40,819)	2.0% (17,948)	3.0% (26,172)	3.9% (34,909)	31.3% (277,737)
DONEGAL	7.0% (62,141)	3.7% (32,455)	4.6% (41,178)	3.5% (30,830)	0.1% (480)	0.0% (0)	0.0% (0)	18.9% (167,084)
MAYO	6.7% (59,138)	3.3% (29,402)	2.8% (24,659)	2.6% (23,444)	0.1% (1,327)	0.0% (0)	0.0% (0)	15.6% (137,970)
CLARE	3.3% (29,140)	3.0% (27,017)	2.9% (26,060)	3.4% (30,192)	0.8% (7,492)	0.7% (6,563)	0.2% (1,474)	14.4% (127,938)
ROSCOMMON	3.4% (29,737)	1.3% (11,937)	1.3% (11,143)	1.6% (14,366)	0.2% (2,072)	0.1% (1,004)	0.0% (0)	7.9% (70,259)
SLIGO	1.1% (9,461)	1.3% (11,714)	1.7% (15,053)	2.0% (17,502)	1.0% (8,512)	0.7% (6,262)	0.2% (1,694)	7.9% (70,198)
LEITRIM	1.3% (11,655)	0.8% (7,264)	0.8% (7,173)	1.0% (9,107)	0.0% (0)	0.0% (0)	0.0% (0)	4.0% (35,199)
Total	30.7% (272,149)	18.1% (160,180)	19.4% (171,887)	18.8% (166,260)	4.3% (37,831)	4.5% (40,001)	4.3% (38,077)	100.0% (886,385)

As shown in Table 2, the largest shares of the total population of the west of Ireland are the 70,877, 62,141 and 59,138 people living in County Galway, County Donegal and County Mayo in Small Areas with no transit at all.

Table 3: SI score & transit supply category by County, weighted by population.

Characteristic	CLARE N = 127,938	DONEGAL N = 167,084	GALWAY N = 277,737	LEITRIM N = 35,199	MAYO N = 137,970	ROSCOMMON N = 70,259	SLIGO N = 70,198
SI							
Min	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Q1	0.1	0.0	0.0	0.0	0.0	0.0	0.9
Mean	183.5	24.5	739.3	32.5	30.3	71.0	323.0
Q3	103.2	24.0	642.7	41.1	13.1	37.1	396.4
Max	3,096.1	684.6	11,860.5	342.6	776.7	1,320.1	3,237.0
Sum	23,471,559.2	4,098,119.7	205,330,898.2	1,145,719.8	4,183,477.4	4,988,734.5	22,671,969.6
transit_supply							
Zero Supply	29,140 (23%)	62,141 (37%)	70,877 (26%)	11,655 (33%)	59,138 (43%)	29,737 (42%)	9,461 (13%)
Very Low	27,017 (21%)	32,455 (19%)	40,391 (15%)	7,264 (21%)	29,402 (21%)	11,937 (17%)	11,714 (17%)
Low	26,060 (20%)	41,178 (25%)	46,621 (17%)	7,173 (20%)	24,659 (18%)	11,143 (16%)	15,053 (21%)
Below average	30,192 (24%)	30,830 (18%)	40,819 (15%)	9,107 (26%)	23,444 (17%)	14,366 (20%)	17,502 (25%)
Above average	7,492 (5.9%)	480 (0.3%)	17,948 (6.5%)	0 (0%)	1,327 (1.0%)	2,072 (2.9%)	8,512 (12%)
High	6,563 (5.1%)	0 (0%)	26,172 (9.4%)	0 (0%)	0 (0%)	1,004 (1.4%)	6,262 (8.9%)
Very High	1,474 (1.2%)	0 (0%)	34,909 (13%)	0 (0%)	0 (0%)	0 (0%)	1,694 (2.4%)

¹ n (%)

There is a statistically significant variation between Counties in the share of Small Areas in each transit supply category ($\chi^2(36) = 700.76, p < .001$). Table 3 indicates that Mayo and Roscommon have higher shares of their population living in Small Areas with no transit supply (over two people in every five), whereas Sligo has the lowest (close to one in eight). Close to one-third of people living in Sligo live in Small Areas with SI scores above the average (not including areas with no transit), followed by a little under one in four in Clare and around one in seven in Roscommon. At the other end of the spectrum, only around one in twelve people in Mayo and Leitrim live in areas with above the average SI scores, while for Donegal it is only one in twenty.

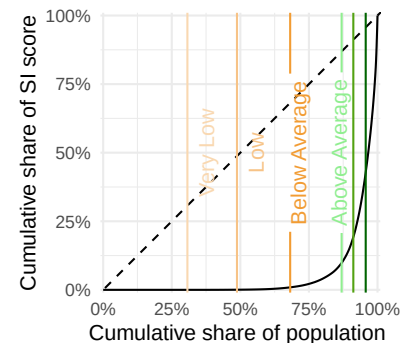


Figure 2: West of Ireland: Lorenz curves

A Lorenz Curve of distribution of the west of Ireland's transit supply to the population is shown in Figure 2. The Gini co-efficient is 0.89, indicating a relatively unequal distribution. In total, the SI score across all of the west of Ireland is approximately 1,030,000, but 99% of this goes to Small Areas that are home to just 32% of the total population. 75% of the population live in Small Areas that, in combination, receive just 3% of the total supply. This is perhaps not surprising, given that providing transit services in rural and low density areas is challenging.

Figure 3 shows Lorenz curves for each of the Counties. Gini co-efficients range from 0.76 for Sligo, through to 0.87 for Roscommon. This suggests that there is some variation, but in general transit supply is unequally distributed to the population across all of the Counties in the west of Ireland.

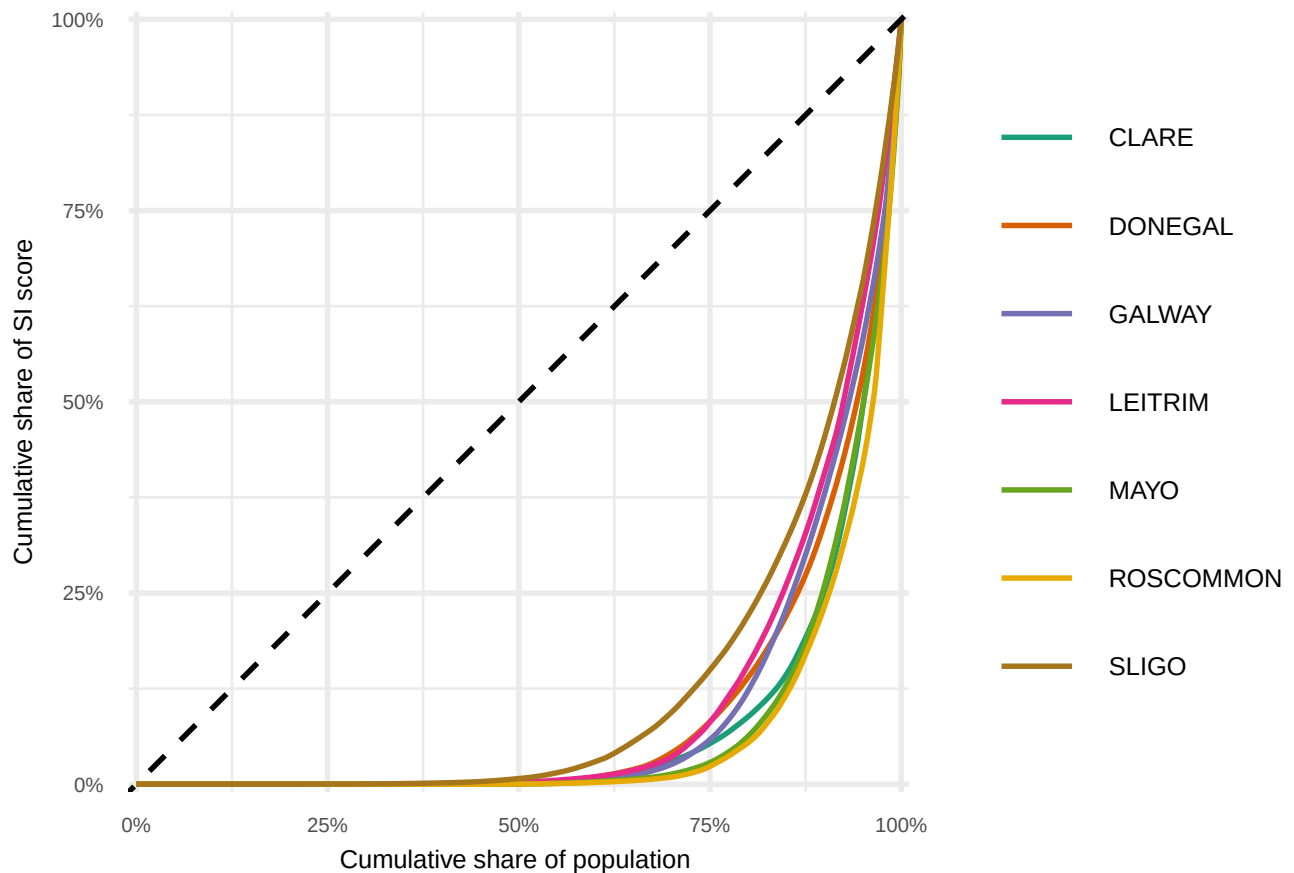


Figure 3: West of Ireland: Lorenz curves by County

Urban areas Approximately 50% of the west of Ireland's population (close to 435,000 people) live in Small Areas that are within one of the 242 named 'urban areas' within the west. Figure 4 shows the total population and aggregate SI scores for each urban area, while Figure

?? shows the same data, but grouped by County

Boxplots of the SI scores for Small Areas within each of the various urban areas are shown in Figure 6, again grouped by County.

County Sligo

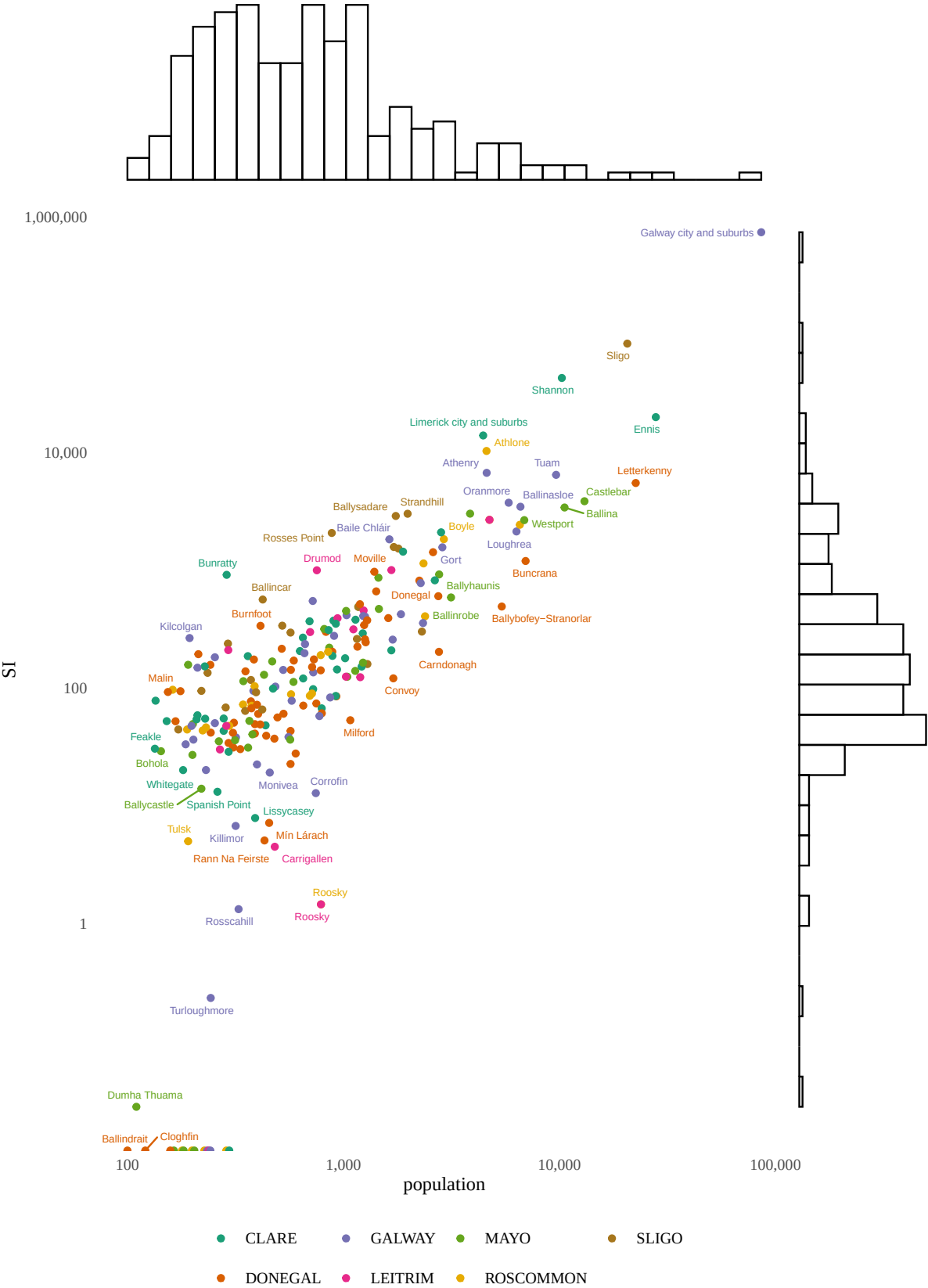


Figure 4: West of Ireland pop. vs SI, aggregated by urban areas



Figure 5: West of Ireland pop. vs SI, aggregated by urban areas

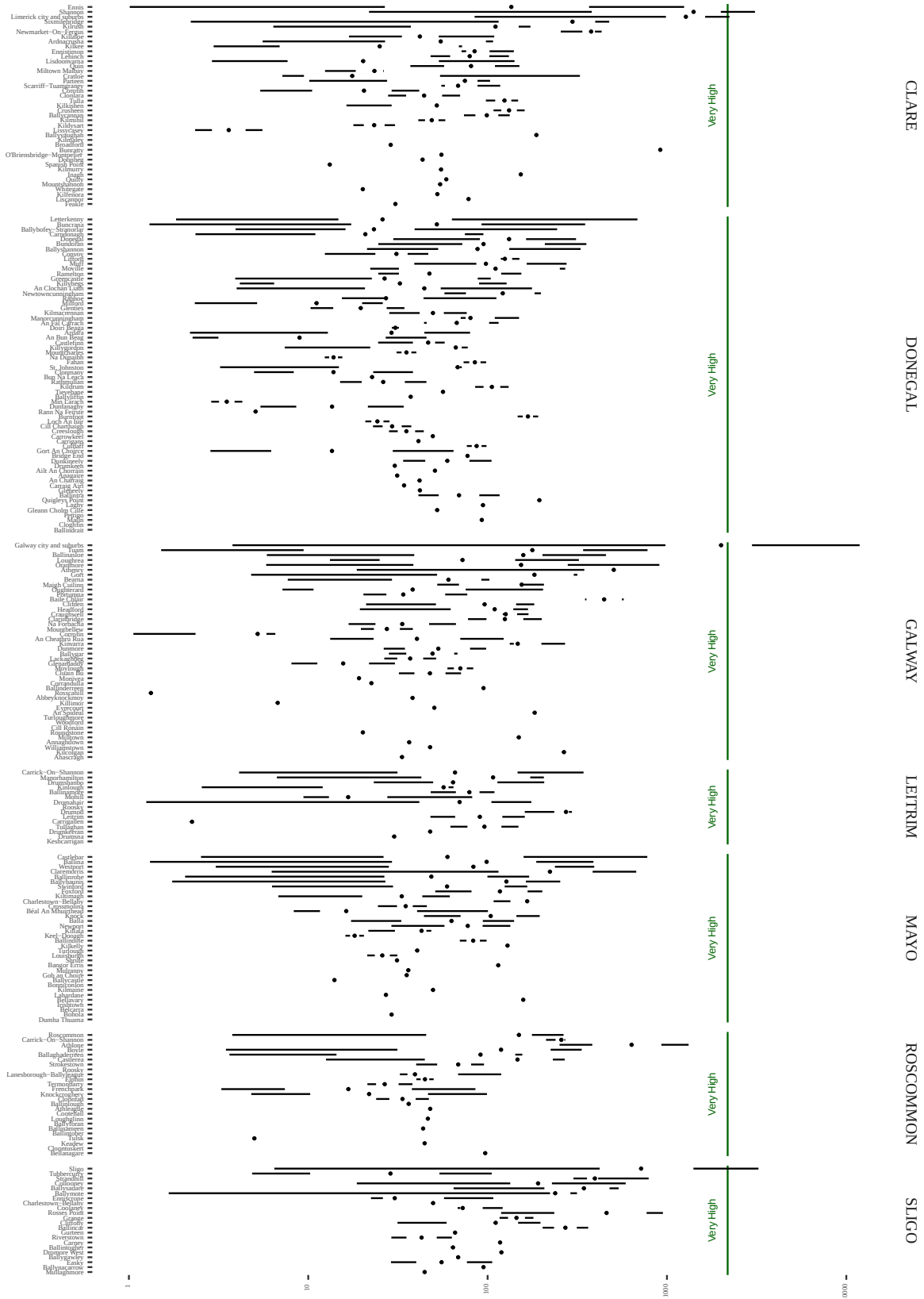


Figure 6: West of Ireland urban areas:
boxplot of Small Area SI scores, ordered
by population size

Table X shows summary statistics for SI scores and the population, overall for all of the County Sligo and split by transit supply category. The total population of County Sligo is 70,198, of whom 77% (53,730) live in Small Areas where the SI score is below the average (of those Small Areas with a non-zero SI score) (413.1). This includes the 9,461 people (13%) who live in Small Areas with no transit at all.

Figure 7 shows transit Supply Index (SI) categories by Small Areas across County Sligo. It also shows to the population living in Small Areas in each category for both the entire County and for various urban areas. In general, there appear to be large areas of County Sligo with SI scores below the average, or with no transit supply at all. Also, as might be expected, Small Areas where SI scores are higher than the average (of those with non-zero scores) appears to be located in very small and scattered geographic areas (at population centers).

Table 2 summarises the Small Area populations by Urban Area in County Sligo by transit supply category, shown as a share of the total County population. Table 3 shows SI scores and transit supply categories by Urban Area, weighted by population.

Table 1: Summary statistics; SI score by transit supply category, by Small Area.

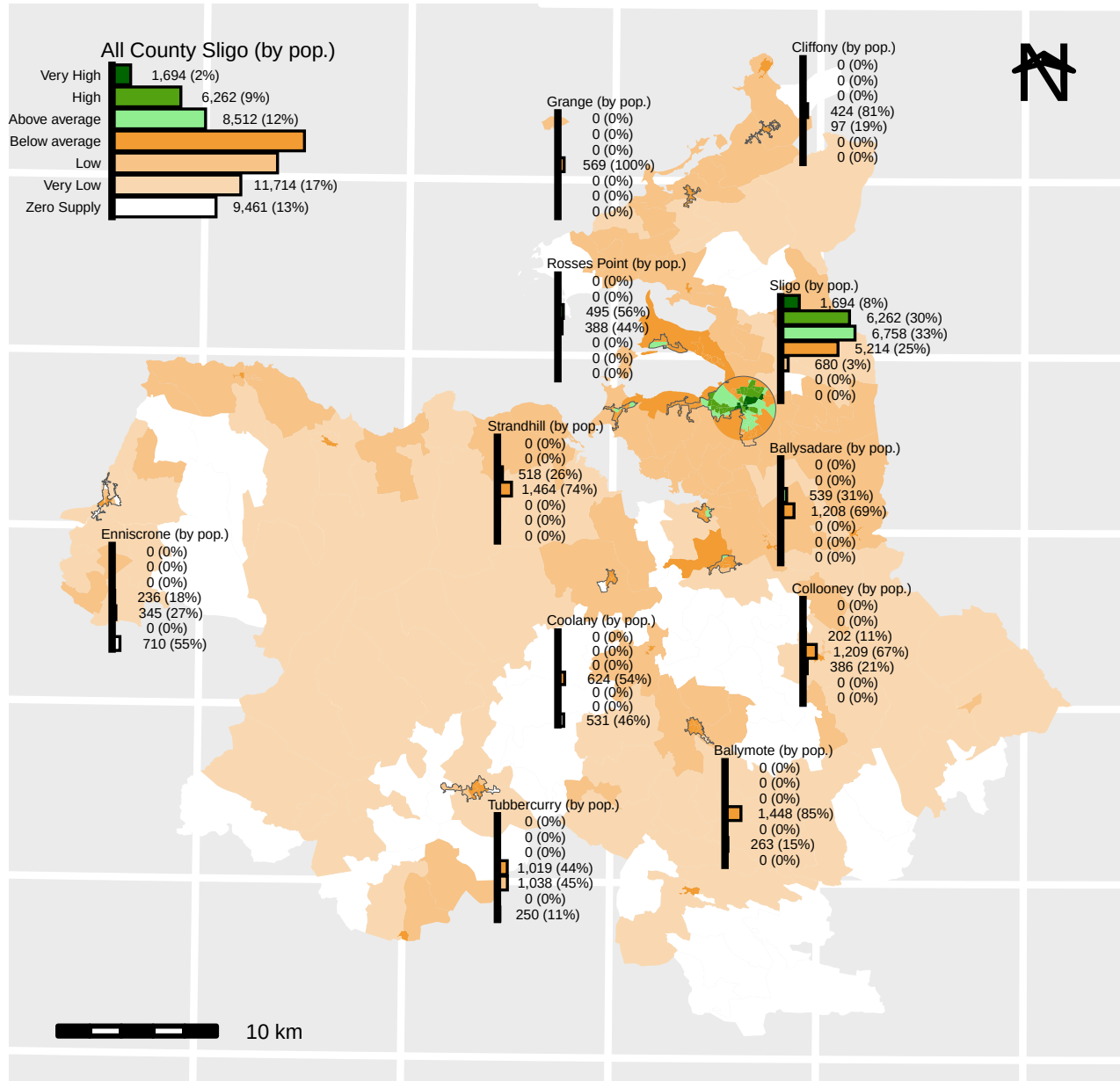


Figure 7: Transit Supply Index categories across County Sligo, by Small Area and by population (for urban areas with > 500 people)

Table 2: County Sligo: Urban Area Small Area populations by Transit Supply grouping

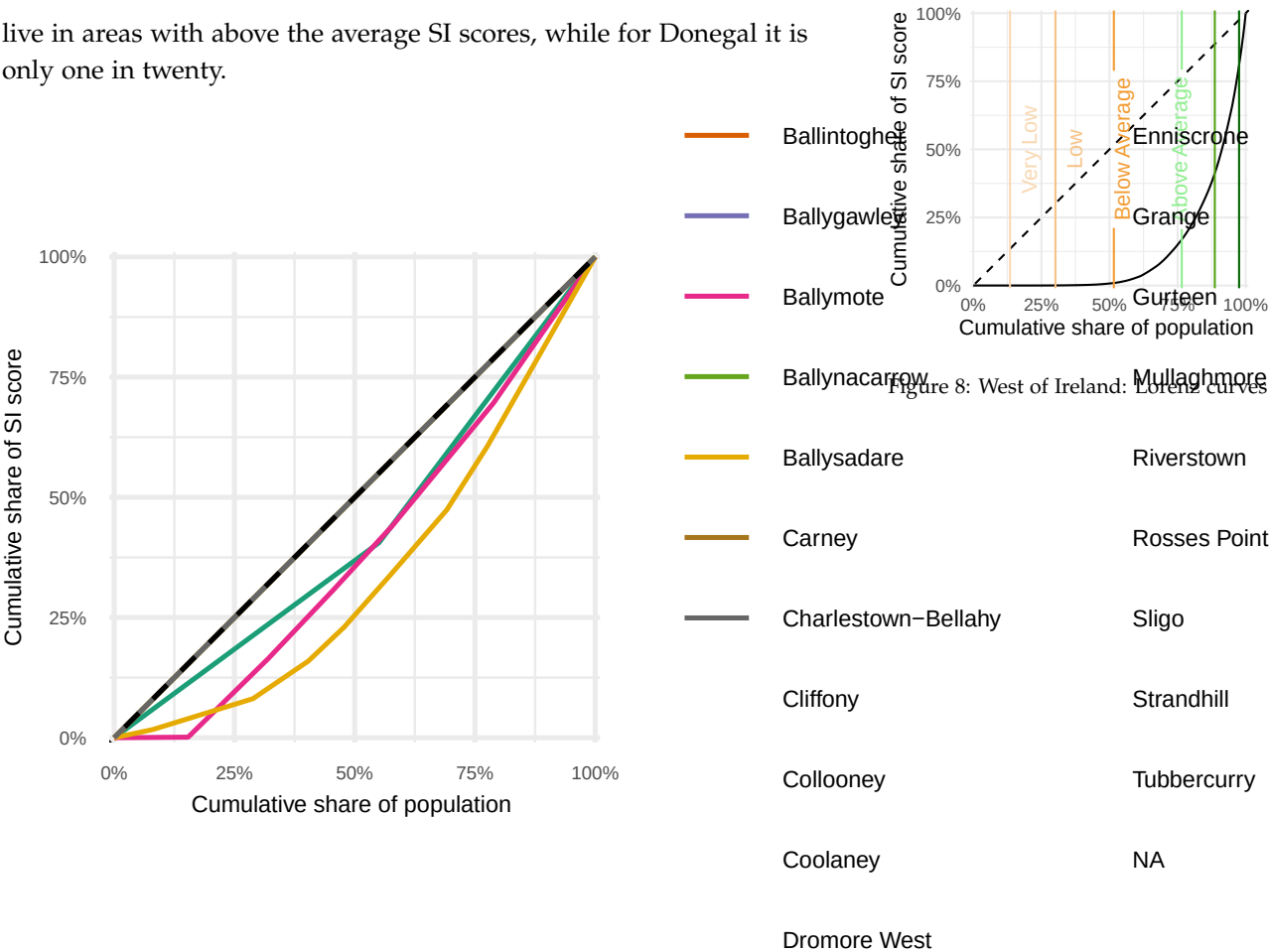
Table 2: County Sligo Urban Area Small Area populations, by transit supply category.

Urban Area	Transit Supply category							Total
	Zero Supply	Very Low	Low	Below av.	Above av.	High	Very High	
Sligo	0.0% (0)	0.0% (0)	1.8% (680)	13.7% (5,214)	17.8% (6,758)	16.5% (6,262)	4.5% (1,694)	54.3% (20,608)
Tubbercurry	0.7% (250)	0.0% (0)	2.7% (1,038)	2.7% (1,019)	0.0% (0)	0.0% (0)	0.0% (0)	6.1% (2,307)
Strandhill	0.0% (0)	0.0% (0)	0.0% (0)	3.9% (1,464)	1.4% (518)	0.0% (0)	0.0% (0)	5.2% (1,982)
Collooney	0.0% (0)	0.0% (0)	1.0% (386)	3.2% (1,209)	0.5% (202)	0.0% (0)	0.0% (0)	4.7% (1,797)
Ballysadare	0.0% (0)	0.0% (0)	0.0% (0)	3.2% (1,208)	1.4% (539)	0.0% (0)	0.0% (0)	4.6% (1,747)
Ballymote	0.0% (0)	0.7% (263)	0.0% (0)	3.8% (1,448)	0.0% (0)	0.0% (0)	0.0% (0)	4.5% (1,711)
Enniscrone	1.9% (710)	0.0% (0)	0.9% (345)	0.6% (236)	0.0% (0)	0.0% (0)	0.0% (0)	3.4% (1,291)
Coolaney	1.4% (531)	0.0% (0)	0.0% (0)	1.6% (624)	0.0% (0)	0.0% (0)	0.0% (0)	3.0% (1,155)
Rosses Point	0.0% (0)	0.0% (0)	0.0% (0)	1.0% (388)	1.3% (495)	0.0% (0)	0.0% (0)	2.3% (883)
Grange	0.0% (0)	0.0% (0)	0.0% (0)	1.5% (569)	0.0% (0)	0.0% (0)	0.0% (0)	1.5% (569)
Cliffony	0.0% (0)	0.0% (0)	0.3% (97)	1.1% (424)	0.0% (0)	0.0% (0)	0.0% (0)	1.4% (521)
Ballincar	0.0% (0)	0.0% (0)	0.0% (0)	1.1% (423)	0.0% (0)	0.0% (0)	0.0% (0)	1.1% (423)
Gurteen	0.4% (155)	0.0% (0)	0.0% (0)	0.7% (266)	0.0% (0)	0.0% (0)	0.0% (0)	1.1% (421)
Riverstown	0.0% (0)	0.0% (0)	0.5% (197)	0.5% (197)	0.0% (0)	0.0% (0)	0.0% (0)	1.0% (394)
Carney	0.0% (0)	0.0% (0)	0.0% (0)	1.0% (373)	0.0% (0)	0.0% (0)	0.0% (0)	1.0% (373)
Ballintogher	0.0% (0)	0.0% (0)	0.0% (0)	0.9% (351)	0.0% (0)	0.0% (0)	0.0% (0)	0.9% (351)
Dromore	0.0% (0)	0.0% (0)	0.0% (0)	0.8% (292)	0.0% (0)	0.0% (0)	0.0% (0)	0.8% (292)
West								
Ballygawley	0.0% (0)	0.0% (0)	0.0% (0)	0.8% (285)	0.0% (0)	0.0% (0)	0.0% (0)	0.8% (285)
Easky	0.0% (0)	0.0% (0)	0.4% (154)	0.2% (81)	0.0% (0)	0.0% (0)	0.0% (0)	0.6% (235)
Ballynacarrow	0.0% (0)	0.0% (0)	0.0% (0)	0.6% (220)	0.0% (0)	0.0% (0)	0.0% (0)	0.6% (220)
Charlestown-Bellahy	0.0% (0)	0.0% (0)	0.0% (0)	0.5% (196)	0.0% (0)	0.0% (0)	0.0% (0)	0.5% (196)
Mullaghmore	0.0% (0)	0.0% (0)	0.0% (0)	0.5% (172)	0.0% (0)	0.0% (0)	0.0% (0)	0.5% (172)
Total	4.3% (1,646)	0.7% (263)	7.6% (2,897)	43.9% (16,659)	22.4% (8,512)	16.5% (6,262)	4.5% (1,694)	100.0% (37,933)

Table 3: County Sligo: SI score & transit supply category by Urban Area, weighted by population.

There is a statistically significant variation between Counties in the share of Small Areas in each transit supply category ($\chi^2(132) = 402.61, p < .001$). Table 3 indicates that Mayo and Roscommon have higher shares of their population living in Small Areas with no transit supply (over two people in every five), whereas Sligo has the lowest (close to one in eight). Close to one-third of people living in Sligo live in Small Areas with SI scores above the average (not including areas with no transit), followed by a little under one in four in Clare and around one in seven in Roscommon. At the other end of the spectrum, only around one in twelve people in Mayo and Leitrim

live in areas with above the average SI scores, while for Donegal it is only one in twenty.



Urban areas

Approximately 50% of County Sligo's population (close to 38,000 people) live in Small Areas that are within one of the 22 named 'urban areas' within County Sligo. Figure 10 shows the total population and aggregate SI scores for each urban area.

Boxplots of the SI scores for Small Areas within each of the various urban areas are shown in Figure 6, again grouped by County.

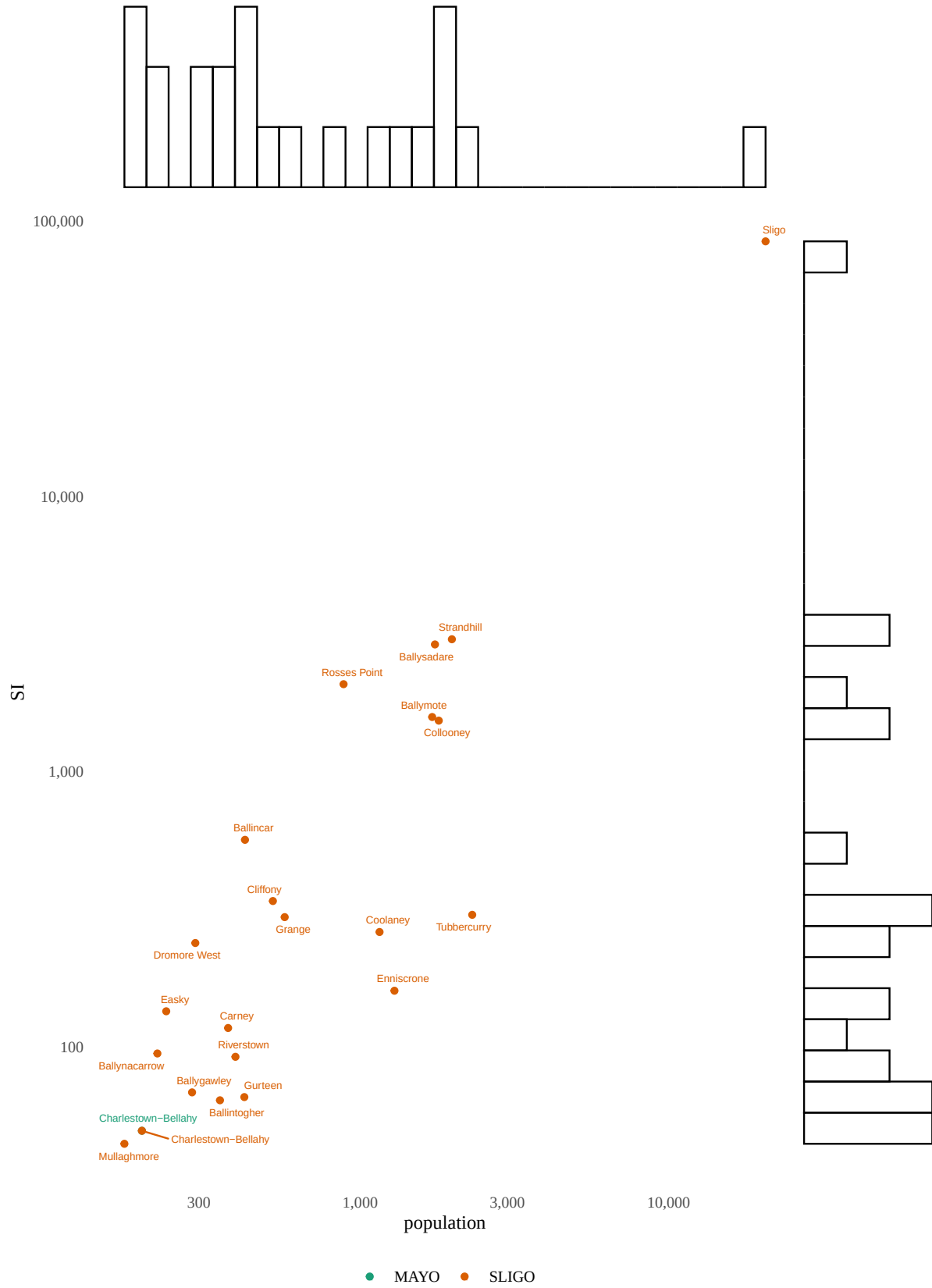


Figure 10: West of Ireland pop. vs SL, aggregated by urban areas

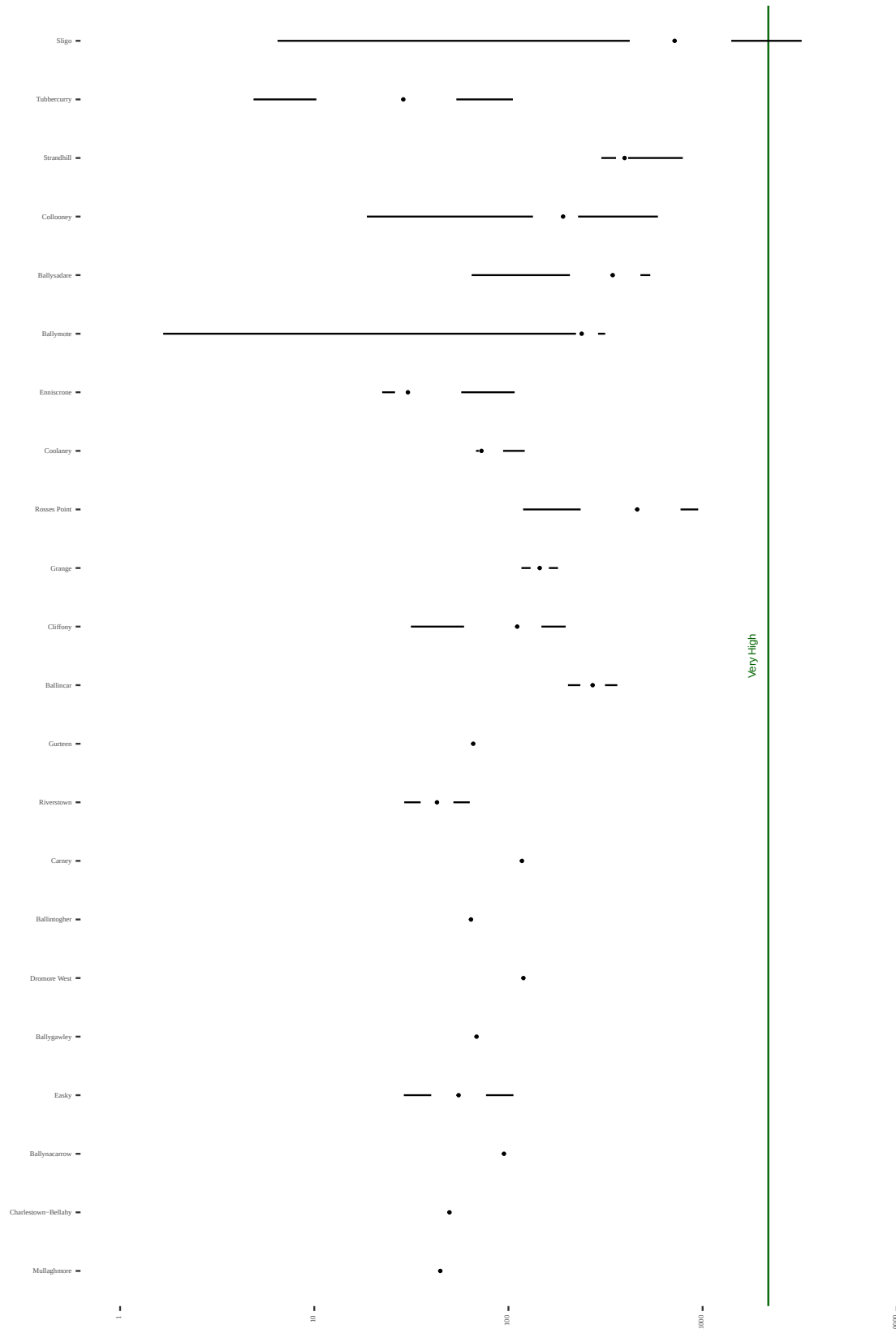


Figure 11: West of Ireland urban areas: boxplot of Small Area SI scores, ordered by population size